

Reduction of order

You are given a D.E and y_1 (1st solution) then you are to find the 2nd solution, y_2 . You will use substitution $y_2 = uy_1$. Plug in y_1 into y_2 and take the Product rule then plug into the equation.

ex:

$$y'' - 10y' + 25y = 0; \quad y_1 = e^{5x}$$

$$y_2 = uy_1 \rightarrow y_2 = ue^{5x}$$

$$y_2 = ue^{5x}$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$f = u$$

$$g = e^{5x}$$

$$\frac{df}{dx} = u'$$

$$\frac{dg}{dx} = 5e^{5x}$$

$$y_2' = u'e^{5x} + 5ue^{5x}$$



$$f = u'$$

$$g = e^{5x}$$

$$\frac{df}{dx} = u''$$

$$\frac{dg}{dx} = 5e^{5x}$$

$$\Rightarrow u''e^{5x} + 5u'e^{5x}$$

$$f = 5u$$

$$g = e^{5x}$$

$$\frac{df}{dx} = 5u'$$

$$\frac{dg}{dx} = 5e^{5x}$$

$$\Rightarrow 5u'e^{5x} + 25ue^{5x}$$

$$y_2'' = u''e^{5x} + 5u'e^{5x} + 5u'e^{5x} + 25ue^{5x}$$

$$y_2'' = u''e^{5x} + 10u'e^{5x} + 25ue^{5x}$$

• We took the first derivative and the second derivative because in the equation we have y' and y'' .

We will now take y_2' and y_2'' and plug into the D.E

$$y_2' = u'e^{5x} + 5ue^{5x}$$

$$y_2'' = u''e^{5x} + 10u'e^{5x} + 25ue^{5x}$$

• for y we will use

$$y_2 = ue^{5x}$$

$$(u''e^{5x} + 10u'e^{5x} + 25ue^{5x}) - 10(u'e^{5x} + 5ue^{5x}) + 25(ue^{5x}) = 0$$

$$u''e^{5x} + \cancel{10u'e^{5x}} + 25ue^{5x} - \cancel{10u'e^{5x}} - \cancel{50ue^{5x}} + 25ue^{5x} = 0$$

$$u''e^{5x} + \cancel{25ue^{5x}} - \cancel{50ue^{5x}} + \cancel{25ue^{5x}} = 0$$

$$\begin{array}{c} u''e^{5x} = 0 \\ \swarrow \quad \searrow \\ \cancel{e^{5x}} = 0 \quad u'' = 0 \end{array}$$

$$\int u'' = \int 0$$

$$u' = C_1$$

$$\int u' = \int C_1$$

$$u = C_1x + C_2$$

Recall: we said $y_2 = u y_1$ you can use this equation to find y_2

$$y_2 = (C_1x + C_2)e^{5x}$$

$$\boxed{y_2 = xe^{5x}}$$

$$y_2 = C_1xe^{5x} + C_2e^{5x}$$

↑
this is
 y_2

↑
this is
 y_1

1)

$$6y'' + y' - y = 0 \quad ; \quad y_1 = e^{x/3}$$

$$y_2 = u y_1 \quad \rightarrow \quad y_2 = u e^{x/3}$$

$$y_2 = u e^{x/3}$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx} \quad f = u \quad g = e^{x/3}$$

$$\frac{df}{dx} = u' \quad \frac{dg}{dx} = \frac{1}{3} e^{x/3}$$

$$y_2' = u' e^{x/3} + \frac{u}{3} e^{x/3}$$

$$f = u'$$

$$g = e^{x/3}$$

$$f = \frac{1}{3} u$$

$$g = e^{x/3}$$

$$\frac{df}{dx} = u''$$

$$\frac{dg}{dx} = \frac{1}{3} e^{x/3}$$

$$\frac{df}{dx} = \frac{1}{3} u'$$

$$\frac{dg}{dx} = \frac{1}{3} e^{x/3}$$

$$\Rightarrow u'' e^{x/3} + \frac{1}{3} u' e^{x/3}$$

$$\Rightarrow \frac{1}{3} u' e^{x/3} + \frac{1}{9} u e^{x/3}$$

$$y_2'' = u'' e^{x/3} + \frac{1}{3} u' e^{x/3} + \frac{1}{3} u' e^{x/3} + \frac{1}{9} u e^{x/3}$$

$$y_2'' = u'' e^{x/3} + \frac{2}{3} u' e^{x/3} + \frac{1}{9} u e^{x/3}$$

$$y_2' = u' e^{x/3} + \frac{u}{3} e^{x/3}$$

• Plug these into the equation

$$y_2'' = u'' e^{x/3} + \frac{2}{3} u' e^{x/3} + \frac{1}{9} u e^{x/3}$$

$$6(u''e^{x/3} + \frac{2}{3}u'e^{x/3} + \frac{1}{9}ue^{x/3}) + (u'e^{x/3} + \frac{4}{3}e^{x/3}) - (ue^{x/3}) = 0$$

$$6u''e^{x/3} + \underline{4u'e^{x/3}} + \cancel{\frac{2}{3}ue^{x/3}} + \underline{u'e^{x/3}} + \cancel{\frac{4}{3}e^{x/3}} - \cancel{ue^{x/3}} = 0$$

$$6u''e^{x/3} + 5u'e^{x/3} = 0 \Rightarrow e^{x/3} (6u'' + 5u') = 0$$

$$\cancel{e^{x/3}} = 0$$

$$6u'' + 5u' = 0$$

$$6u'' + 5u' = 0$$

$$u'' + \frac{5}{6}u' = 0$$

$$\text{let } w = u' \\ \Rightarrow w' = u''$$

• when you end up with something like this you will most likely do a linear equation

$$w' + \frac{5}{6}w = 0 \Rightarrow \frac{dw}{dx} + \frac{5}{6}w = 0$$

$$p(x) = \frac{5}{6}, \quad e^{\int \frac{5}{6} dx} \Rightarrow e^{\frac{5}{6}x} \leftarrow \text{this is called the integrating factor}$$

$$\frac{d}{dx} \left[\begin{matrix} \text{integrating} \\ \text{factor} \end{matrix} \cdot w \right] = \text{right of equation}$$

$$\frac{d}{dx} [e^{\frac{5}{6}x} \cdot w] = 0$$

$$\int \frac{d}{dx} [e^{\frac{5}{6}x} \cdot w] = \int 0$$

$$e^{\frac{5}{6}x} \cdot w = C_1$$

$$w = c_1 e^{-\frac{5}{6}x}$$

recall we said $w = u'$

$$\int u' = \int c_1 e^{-\frac{5}{6}x}$$

$$u = -\frac{5}{6} c_1 e^{-\frac{5}{6}x} + c_2$$

recall we said $y_2 = u y_1$

$$y_2 = \left(-\frac{5}{6} c_1 e^{-\frac{5}{6}x} + c_2 \right) e^{x/3}$$

$$y_2 = -\frac{5}{6} c_1 e^{-\frac{x}{2}} + c_2 e^{x/3}$$

$\uparrow$
 y_2

$\uparrow$
 y_1

$y_2 = e^{-x/2}$

2)

$$y'' + 36y = 0 ; \quad y_1 = \cos(6x)$$

$$y_2 = u y_1 \rightarrow y_2 = u \cos(6x)$$

$$y_2 = u \cos(6x)$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$f = u \quad g = \cos(6x)$$

$$\frac{df}{dx} = u' \quad \frac{dg}{dx} = -6 \sin(6x)$$

$$y_2' \Rightarrow u' \cos(6x) - 6u \sin(6x)$$

$$f = u'$$

$$g = \cos(6x)$$

$$\frac{df}{dx} = u''$$

$$\frac{dg}{dx} = -6 \sin(6x)$$

$$\Rightarrow u'' \cos(6x) - 6u' \sin(6x)$$

$$f = -6u$$

$$g = \sin(6x)$$

$$\frac{df}{dx} = -6u'$$

$$\frac{dg}{dx} = 6 \cos(6x)$$

$$\Rightarrow -6u' \sin(6x) - 36u \cos(6x)$$

$$y_2' = u' \cos(6x) - 6u' \sin(6x) - 6u' \sin(6x) - 36u \cos(6x)$$

$$y_2'' = u'' \cos(6x) - 12u' \sin(6x) - 36u \cos(6x)$$

$$y_2' = u' \cos(6x) - 6u \sin(6x)$$

• Plug these into the equation

$$y_2'' = u'' \cos(6x) - 12u' \sin(6x) - 36u \cos(6x)$$

$$(u''(\cos(6x)) - 12u' \sin(6x) - 36u \cos(6x)) + 36(u \cos(6x)) = 0$$

$$u''(\cos(6x)) - 12u' \sin(6x) - \cancel{36u \cos(6x)} + \cancel{36u \cos(6x)} = 0$$

$$u'' \cos(6x) - 12u' \sin(6x) = 0$$

$$\frac{u''}{u'} = 12 \frac{\sin(6x)}{\cos(6x)} \longrightarrow \frac{u''}{u'} = 12 \tan(6x)$$

$$\text{let } w = u', \quad w' = u''$$

$$\frac{w'}{w} = 12 \tan(6x)$$

$$\int \frac{dw}{w} = \int 12 \tan(6x) dx$$

$$\begin{aligned} u &= 6x \\ du &= 6dx \\ \frac{du}{6} &= dx \end{aligned}$$

$$\ln(w) = \frac{12}{6} \int \tan(u) du$$

$$\ln(w) = 2 \ln|\sec(6x)| + C_1$$

$$\cancel{e^{\ln(w)}} = \cancel{e^{\ln|\sec(6x)|}} + C_1$$

$$\text{let } e^C = C$$

$$w = C \sec^2(6x)$$

$$\text{Recall } w = u'$$

$$\int u' = \int C \sec^2(6x) dx$$

$$\begin{aligned} u &= 6x \\ du &= 6dx \end{aligned}$$

$$u = C \int \sec^2(6x) dx$$

$$\frac{du}{6} = dx$$

$$u = \frac{C}{6} \int \sec(u) du$$

$$u = \frac{C}{6} \tan(6x) + D$$

recall we said $y_2 = u y_1$

$$y_2 = \left(\frac{C}{6} \tan(6x) + D \right) \cos(6x)$$

$$y_2 = \frac{C}{6} \left(\frac{\sin(6x)}{\cos(6x)} \right) \cancel{\cos(6x)} + D \cos(6x)$$

$$y_2 = \frac{C}{6} \sin(6x) + D \cos(6x)$$

$\uparrow$ y_2 $\uparrow$ y_1

$y_2 = \sin(6x)$

3)

$$x^2 y'' - 11x y' + 36y = 0; \quad y_1 = x^6$$

$$y_2 = u y_1 \quad \Rightarrow \quad y_2 = u x^6$$

$$y_2 = u x^6$$

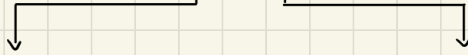
$$\frac{df}{dx} \cdot y + f \cdot \frac{dy}{dx}$$

$$f = u \quad g = x^6$$

$$\frac{df}{dx} = u' \quad \frac{dg}{dx} = 6x^5$$

$$y_2' = u' x^6 + 6u x^5$$

$$y_2' = u'x^6 + 6ux^5$$



$$f = u'$$

$$g = x^6$$

$$\frac{df}{dx} = u''$$

$$\frac{dg}{dx} = 6x^5$$

$$f = 6u$$

$$g = x^5$$

$$\frac{df}{dx} = 6u'$$

$$\frac{dg}{dx} = 5x^4$$

$$\Rightarrow u''x^6 + 6u'x^5$$

$$\Rightarrow 6u'x^5 + 30ux^4$$

$$y_2'' = u''x^6 + 6u'x^5 + 6u'x^5 + 30ux^4$$

$$y_2'' = u''x^6 + 12u'x^5 + 30ux^4$$

$$y_2' = u'x^6 + 6ux^5$$

$$y_2'' = u''x^6 + 12u'x^5 + 30ux^4$$

• Plug these into the equation

$$x^2(u''x^6 + 12u'x^5 + 30ux^4) - 11x(u'x^6 + 6ux^5) + 36(ux^6) = 0$$

$$u''x^8 + \underline{12u'x^7} + \underline{30ux^6} - \underline{11u'x^7} - \underline{66ux^6} + \underline{36ux^6} = 0$$

$$u''x^8 + u'x^7 = 0$$

$$x^7(u''x + u') = 0$$

$$\cancel{x^7} = 0 \quad u''x + u' = 0$$

$$\text{let } w = u' \Rightarrow w' = u''$$

$$w'x + w = 0$$

$$w'x = -w$$

$$\frac{w'}{w} = -\frac{1}{x}$$

$$\int \frac{dw}{w} = \int -\frac{1}{x}$$

$$\ln|w| = -\ln|x| + C$$

$$\cancel{e^{\ln|w|}} = \cancel{e^{\ln|x|}} + C$$

$$\text{let } e^C = c_1$$

$$w = \frac{c_1}{x}$$

recall $w = u'$

$$\int u' = \int \frac{c_1}{x}$$

$$u = c_1 \ln|x| + c_2$$

recall $y_2 = u y_1$

$$y_2 = u y_1 \Rightarrow (c_1 \ln|x| + c_2) x^6$$

$$y_2 = c_1 \ln|x| x^6 + x^6 c_2$$

$\uparrow$ y_2 $\uparrow$ y_1

$y_2 = x^6 \ln|x|$